

The later dimension-constancy theorem for arXiv:1909.00381

Literature-status note for `fa_banach_001`

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Status. `literature_already_answered` (dimension constancy and the resulting one-stratum perimeter formula).

Original open direction

Let $(X, d, \mathfrak{m})$ be an $\text{RCD}(K, N)$ space of essential dimension n , and let $E \subset X$ have finite perimeter. The source decomposes the reduced boundary into strata $\mathcal{F}_k E$, whose blow-ups are k -dimensional Euclidean half-spaces, and proves

$$|D\chi_E| = \sum_{k=1}^n \frac{\omega_{k-1}}{\omega_k} \mathcal{H}^h \llcorner \mathcal{F}_k E. \quad (1)$$

Here $\mathcal{H}^h$ is the codimension-one Hausdorff-type measure with gauge $h(B_r(x)) = \mathfrak{m}(B_r(x))/r$. On source PDF page 4, the authors ask whether tangent dimension is constant for finite-perimeter boundaries and whether (1) can be sharpened using the top-dimensional case [1].

Later answer

The same authors answered this in arXiv:2109.12585. Its introduction explicitly identifies the possibility that $|D\chi_E|(\mathcal{F}_k E) > 0$ for $k < n$ as the question left open in the source, and calls its first main result a negative answer.

Theorem 1.3 on answering PDF page 3 (equivalently Corollary 3.2 on page 16) states that

$$|D\chi_E|(\mathcal{F}_k E) = 0 \quad (k \neq n). \quad (2)$$

It follows that the perimeter measure is concentrated on the ambient n -regular stratum and (1) becomes the single-term formula

$$|D\chi_E| = \frac{\omega_{n-1}}{\omega_n} \mathcal{H}^h \llcorner \mathcal{F}_n E. \quad (3)$$

This is exactly the requested constancy and the corresponding top-dimensional sharpening. The later Theorem 3.1 is stronger: the total variation $|Df|$ of every BV function is concentrated on the n -regular set. Theorem 3.4 also proves that the tangent module over every finite-perimeter boundary has constant dimension n [2].

Scope

The resolution is complete for dimension constancy and the reduction of the source's stratified formula to (3). In a general collapsed space, the measure in (3) remains the reference-measure-dependent $\mathcal{H}^h$. This note therefore does not claim a general conversion to ordinary $(n - 1)$ -dimensional Hausdorff measure. In the non-collapsed case the later paper records that the expected conversion is already available.

References

- [1] E. Brué, E. Pasqualetto, and D. Semola, *Rectifiability of the reduced boundary for sets of finite perimeter over $\text{RCD}(K, N)$ spaces*, arXiv:1909.00381, <https://arxiv.org/abs/1909.00381>.
- [2] E. Brué, E. Pasqualetto, and D. Semola, *Constancy of the dimension in codimension one and locality of the unit normal on $\text{RCD}(K, N)$ spaces*, arXiv:2109.12585, <https://arxiv.org/abs/2109.12585>.